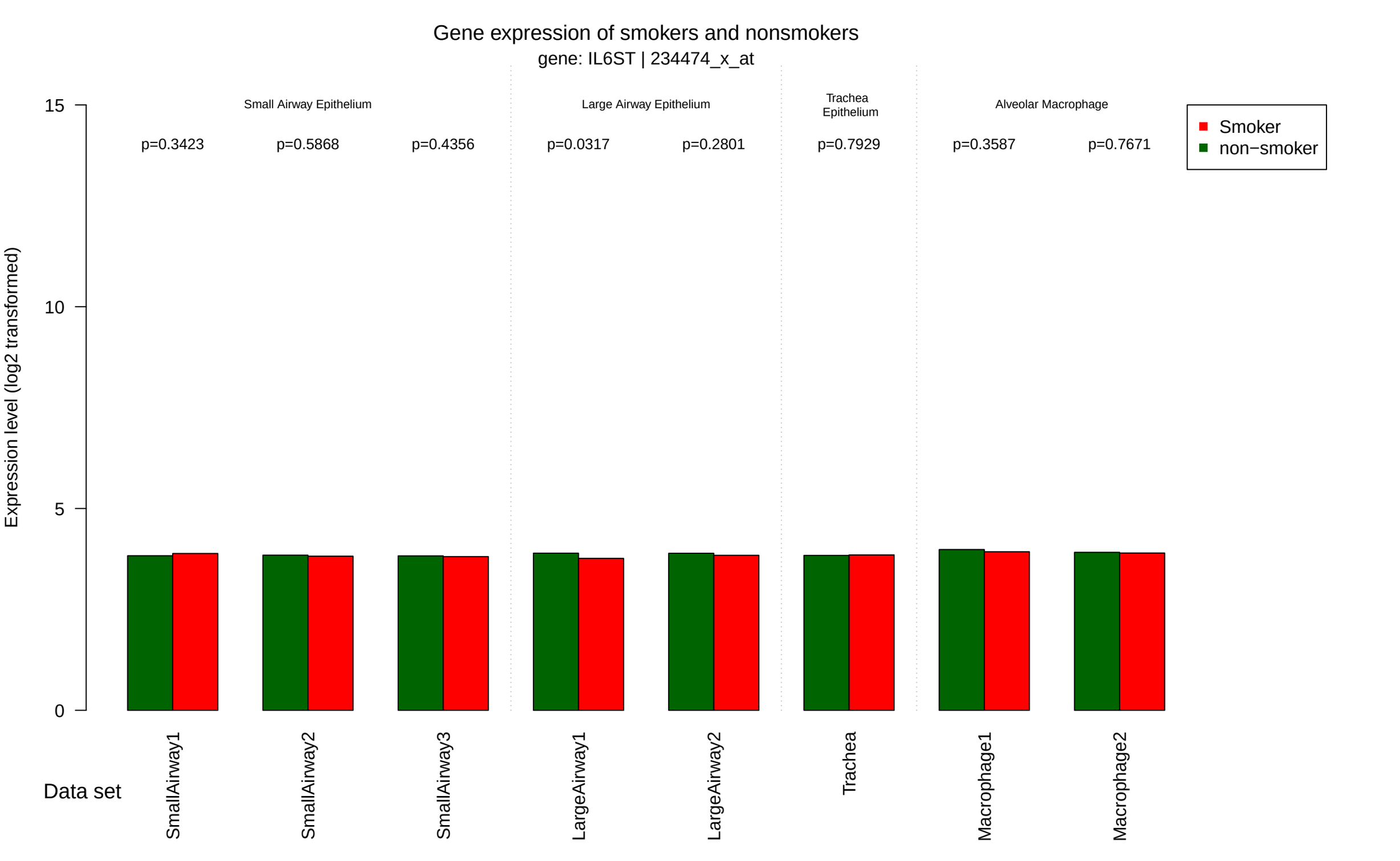
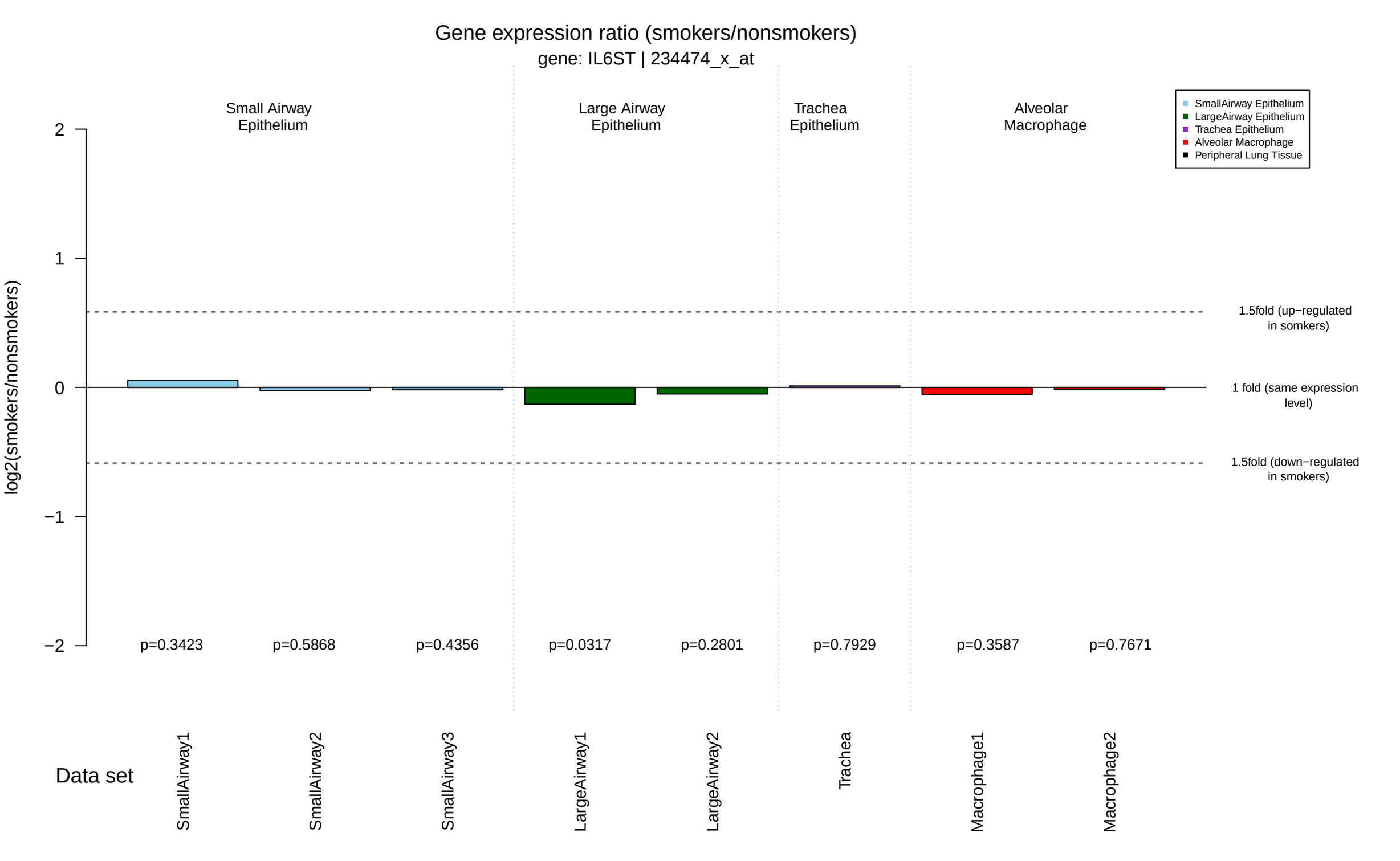
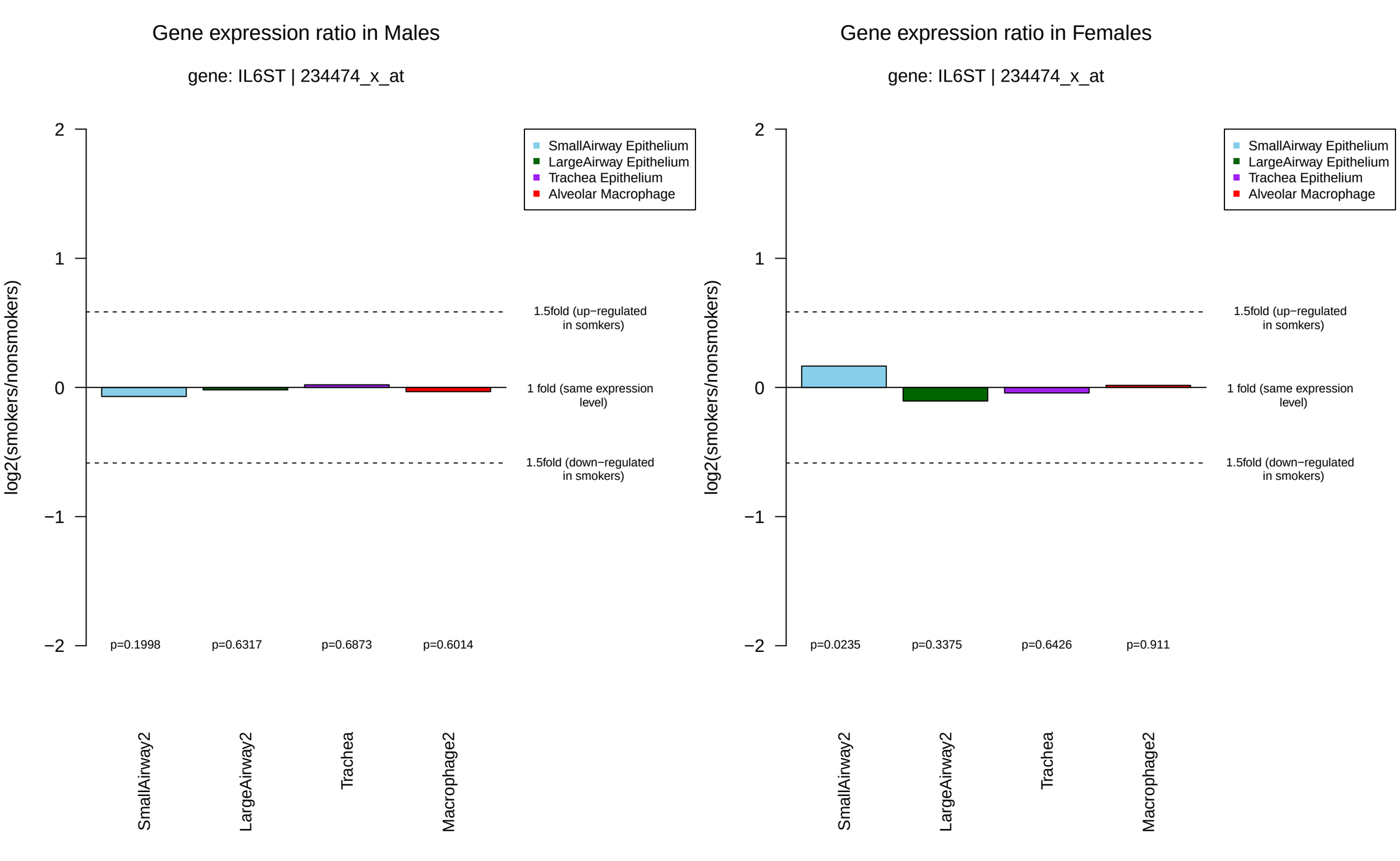
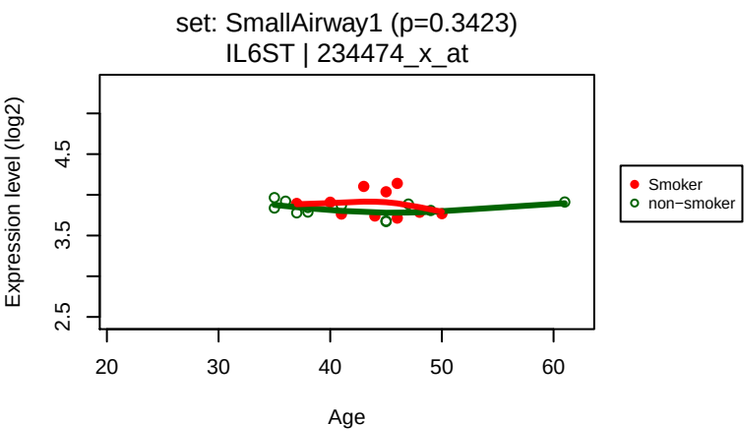
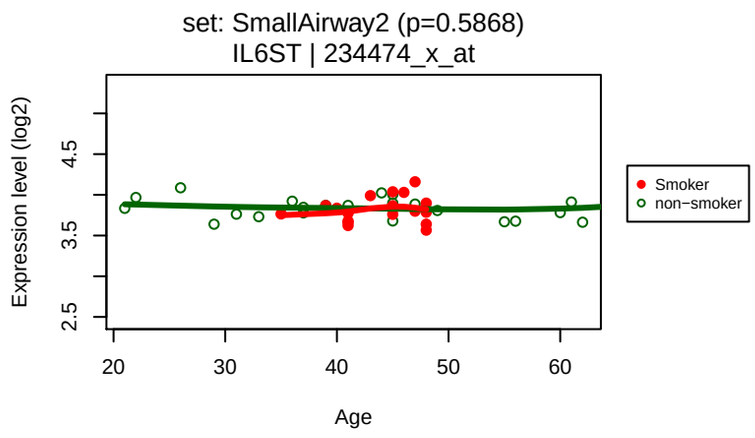
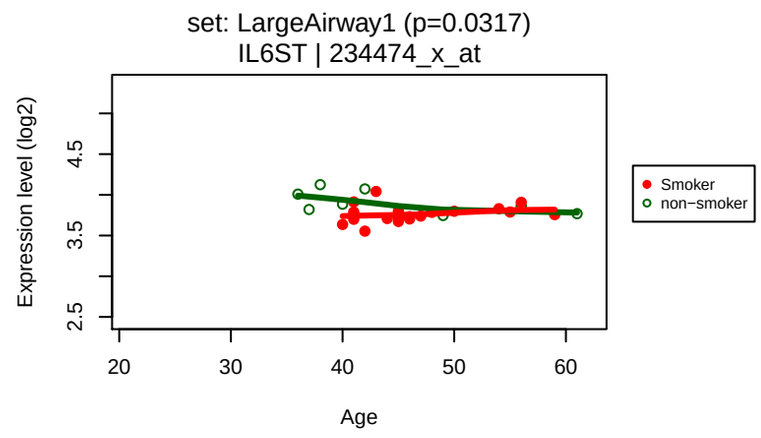

Gene expression of smokers and nonsmokers
gene: IL6ST | 234474_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3423
p=0.5868
p=0.4356
p=0.0317
p=0.2801
p=0.7929
p=0.3587
p=0.7671
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

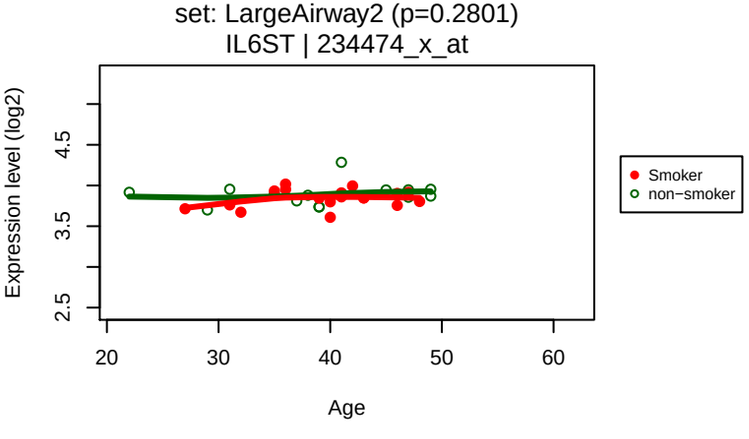

Gene expression ratio (smokers/nonsmokers)
gene: IL6ST | 234474_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3423
p=0.5868
p=0.4356
p=0.0317
p=0.2801
p=0.7929
p=0.3587
p=0.7671
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

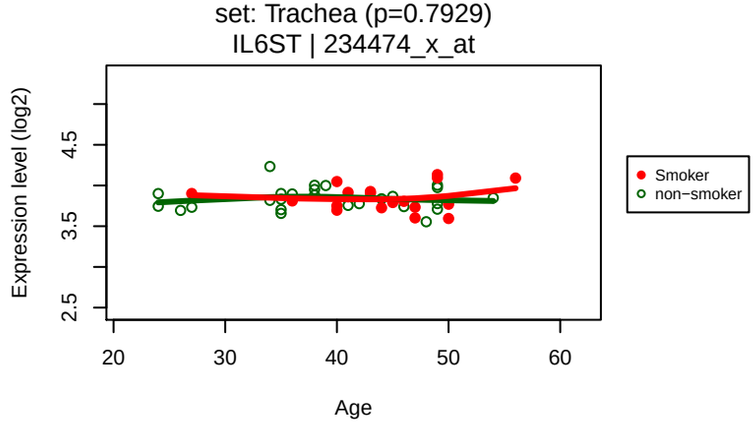

Gene expression ratio in Males
gene: IL6ST | 234474_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1998
p=0.6317
p=0.6873
p=0.6014
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL6ST | 234474_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0235
p=0.3375
p=0.6426
p=0.911
SmallAirway2
LargeAirway2
Trachea
Macrophage2

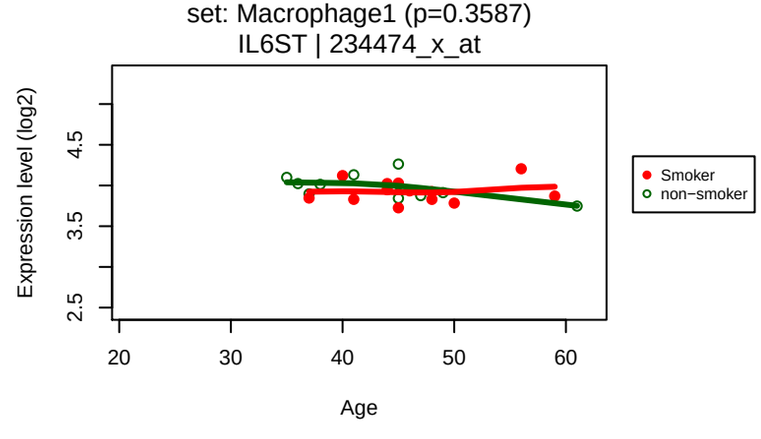

set: SmallAirway1 (p=0.3423)
IL6ST | 234474_x_at
Expression level (log2)
Age
Smoker
non−smoker

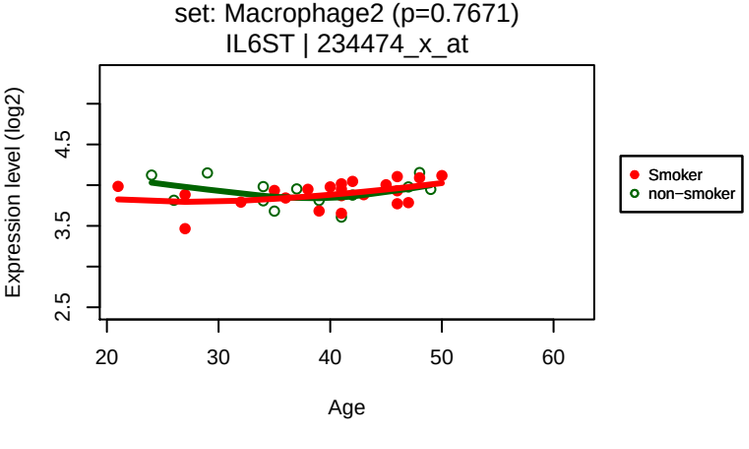

set: SmallAirway2 (p=0.5868)
IL6ST | 234474_x_at
Expression level (log2)
Age
Smoker
non−smoker

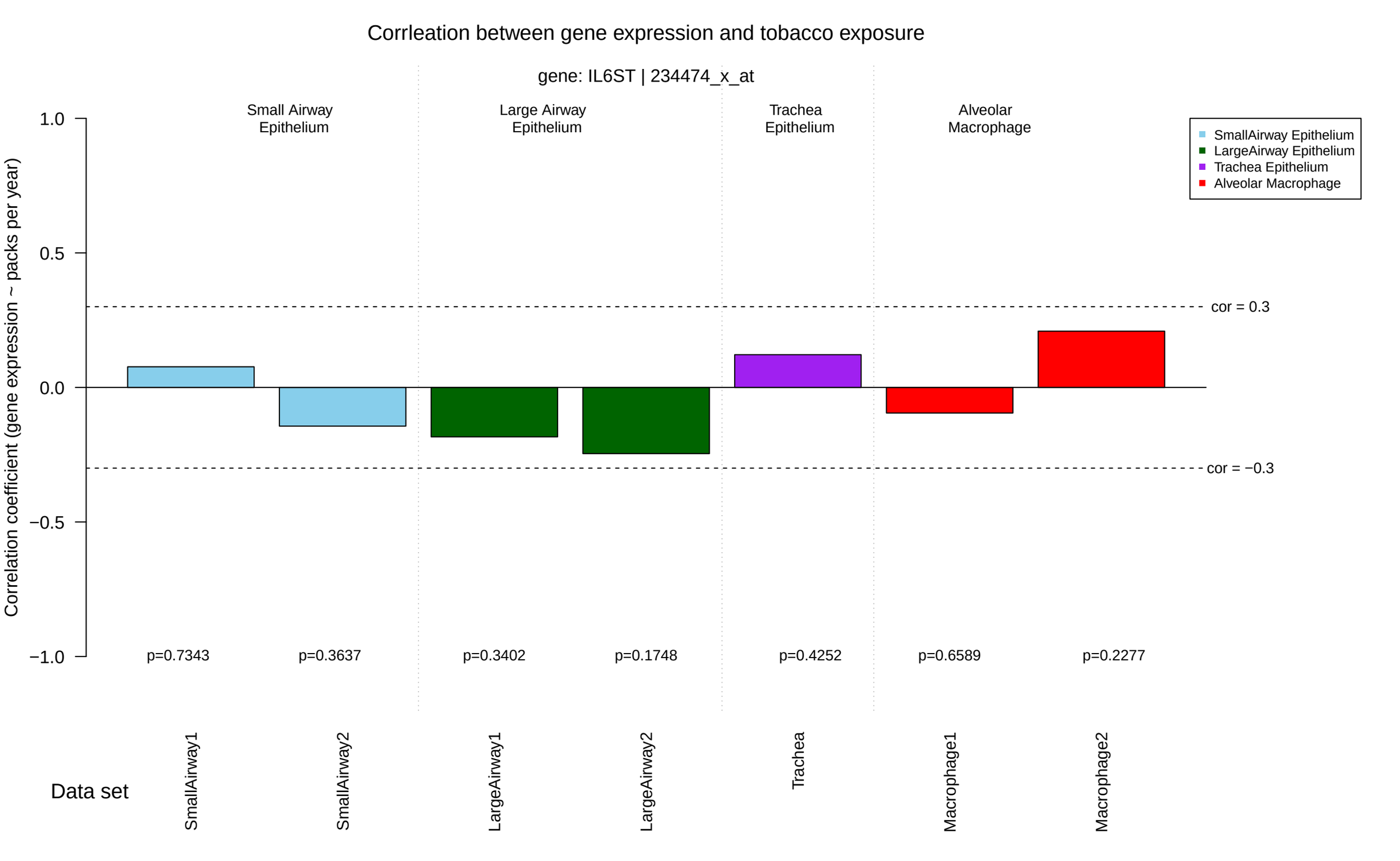

set: LargeAirway1 (p=0.0317)
IL6ST | 234474_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2801)
IL6ST | 234474_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7929)
IL6ST | 234474_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3587)
IL6ST | 234474_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7671)
IL6ST | 234474_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL6ST | 234474_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7343
p=0.3637
p=0.3402
p=0.1748
p=0.4252
p=0.6589
p=0.2277
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2